Paired Fault Assessment of a Fly-Connectome Spiking Controller

Shuchao Gao
Waseda University
Tokyo, Japan

***Abstract*—Task-level fault loss in a spiking controller can depend on feedback even when healthy performance matches. We provide an auditable paired assessment with shared input variates and deletion masks, causal feedback, exact scheduled actuation and independent input/geometry replay. A 138,639-neuron fly-connectome model is tested on 24 new seed clusters crossing two laws, two bearings and three network conditions. All 96 healthy episodes remain collision-free through one second. At 50% connection-row deletion, proportional and relay control succeed in 37/48 and 48/48 episodes; their paired loss difference is 22.9 percentage points, with a 95% cluster interval of 12.5–33.3 and Holm-adjusted $p = 0.00195$. The 25% contrast remains uncertain. A frozen 400-ms command-rate criterion detects only 2/13 later collisions. Geometry-informed forecasting detects all 13 but flags 57/275 successful tasks. Task loss changes with the declared horizon; stopped-duration matching leaves new group activity medians unchanged. These findings support paired task assessment without establishing biological or hardware fault tolerance.**

***Index Terms*—spiking neural networks, fault assessment, feedback control, connectome, neuromorphic systems**

I. Introduction

An embedded spiking controller must produce an effective action before contact. A neural-rate criterion can therefore accept an episode that fails its task, while altered network activity can accompany successful behavior. Assessment requires a defined plant, a causal neural-to-actuator interface and a declared observation horizon.

Closed-loop comparisons also couple faults to future sensory input. Equal seed labels alone need not align random inputs when stopping or event consumption changes. Early termination changes the interval over which spikes are accumulated. Stopped-episode, common-duration and fixed-horizon counts answer different resource questions. Their relationship to task failure should be measured on paired records after verifying healthy capability in every direction.

We couple a fly-connectome spiking substrate to a synthetic pursuit-avoidance task. Its assumed receptive fields and body coupling define an engineering stress model, not biological escape. Separate development and validation establish two bilateral laws; new seed clusters cross law, direction and nested deletion dose at a fixed one-second endpoint.

Our engineering contribution is an executable paired protocol with source-bound registrations, shared input variates and masks, exact actuation and independent input/geometry replay. The prospective experiment demonstrates controller-dependent fault loss despite matched healthy task success, and quantifies errors when neural criteria are used as task-failure warnings. Standard statistical and kinematic components support these bounded findings; they are not claimed as new algorithms. All failed development attempts and test cells are retained.

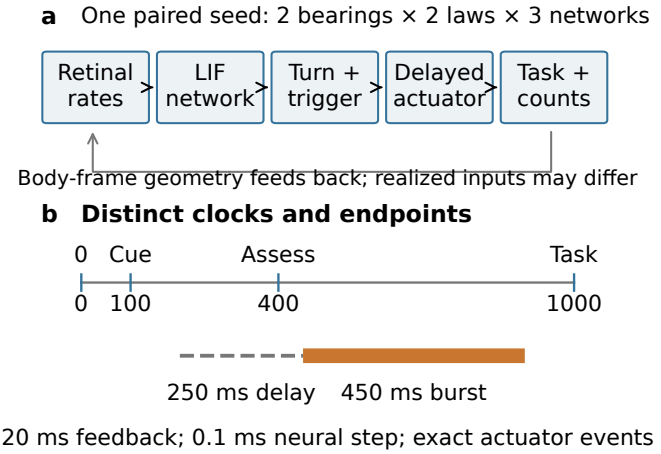


Fig. 1. One seed couples all twelve core conditions. Neural decisions affect the next motion frame; actuator boundaries are exact. The example trigger is 200 ms. Fixed precontact warnings, first contact and resource exposure have different endpoints.

II. Positioning and Scope

Shiu et al. [1] connect anatomical wiring to executable sensorimotor hypotheses. An's sparsification study [2] varies connection strength, placement, state and task; Jo [3] reports neural and virtual-arena responses to connectome damage. These works already establish context- and endpoint-dependent function. Our question concerns source-auditable pairing, sampled-data actuation and same-record exposure estimands in a prospective assessment, rather than a new claim that endpoint choice matters.

SpikeFI [4] and SpikingJET [5] provide SNN injection frameworks with application evaluation. NeuroBench [6] and embodied-agent benchmarking [7] motivate separate functional and resource measurements. Here the comparisons rescore our own generated records; neither injector framework is executed as a benchmark. Physical neutron tests [8] illustrate the additional evidence needed to connect software fault models to hardware. Our weighted-row deletions have no calibrated physical upset rate, power or area interpretation.

III. Paired Closed-Loop Assessment Protocol

A. Substrate and Randomness

The released FlyWire v783 graph has 138,639 neurons and 15,091,983 signed connection-table rows, each representing multiple contacts. We use the Shiu LIF parameterization [1] in Brian 2 [9], NumPy backend, 0.1 ms steps. The original publication used v630. Sensory cells retain 2.2 ms refractoriness, a stated stimulation variant. Initial states and unmodified weights are restored before each episode; equations and parameters accompany the source.

An assumed receptive-field map supplies geometry-dependent rates up to 150 Hz to 104 LC4 and 210 LPLC2 cells. Other external rates are zero. Eight descending cells form a synthetic command interface. Anatomy and looming selectivity [10] motivate the substrate; the map and body coupling are not validated fly behavior. Actual cell identities and map parameters are saved.

Each seed crosses both bearings, both laws and healthy/25%/50% deletion. Shared row variates define $w_j^{(f)} = w_j I(V_{sj} \geq f)$, giving nested masks. All contacts in a row are removed together. Separate frame-indexed uniforms generate inputs $I[U_{sk\tau i} < \lambda_{ki}\Delta t]$. Uniforms are common across conditions; realized events differ through feedback-dependent λ. Digests bind masks, uniforms, actual events, maps, sources and external tables. Seed traceability does not imply cross-platform bit identity.

B. Causal Feedback and Actuation

Body-frame bearing subtracts heading from world-relative bearing. Looming uses changes in $\alpha = \arctan(0.05/d)$ at actual separation d. Input begins at 100 ms; decisions at each frame end affect the *next* 20 ms motion interval.

Twenty passive episodes (four seeds; $-60, -30, 0, 30, 60°$) fit a ridge decoder $z_k = \beta_0 + \boldsymbol{\beta}^\top \bar{\mathbf{r}}_k$ of bearing/60°, using five-frame means of eight command rates. Both laws set $\omega_k = 0$ before the cue or below the 4.03125 Hz eight-cell mean trigger level; otherwise

$$\omega_k = \begin{cases} -600\,\mathrm{clip}(z_k, -1, 1), & \text{proportional}, \\ -600\,\mathrm{sgn}(z_k), & \text{relay}, \end{cases} \quad [\mathrm{deg/s}]. \quad (1)$$

The first qualifying decision t_d schedules one 0.35 m/s heading-fixed burst on $[t_d + 250, t_d + 700)$ ms. Otherwise speed is 0.05 m/s and heading is capped at $\pm 100°$. Motion segments split at exact actuator boundaries; positions use starting heading, updated at each segment end. A 0.3 m/s pursuer starts 0.25 m away and aims at the agent at each segment's beginning. First contact at distance ≤ 0.05 m is solved inside each linear relative-motion segment.

Success means collision-free through 1 s, not eventual escape. Failure latches at first contact; the specified trajectory and neural loop continue for equal 1 s resource exposure. Post-contact activity is counterfactual accounting, not physical collision dynamics or warning data. Independent replay checks input events, geometry, decisions, clocks and exact-horizon state without rerunning neural dynamics.

C. Paired Estimands

Let $S_F(t), S_H(t)$ be paired network counts, and T_F, T_H their contact-frame end or 1 s. Distinguish

$$R_{\mathrm{stop}} = \frac{S_F(T_F)}{S_H(T_H)}, \quad R_{\mathrm{common}} = \frac{S_F(T_*)}{S_H(T_*)}, \quad T_* = \min(T_F, T_H). \quad (2)$$

Fixed 400 ms and 1 s ratios are separate estimands. The former precedes all contacts; common duration remains outcome-dependent and does not equate sensory histories. For binary outcomes, report loss $Y_H(1 - Y_F)$, gain $(1 - Y_H)Y_F$ and net loss $D = Y_H - Y_F$. The controller interaction is $D_{\mathrm{relay}} - D_{\mathrm{proportional}}$, averaging bearings within seed.

D. Validation, Warnings and Inference

An initial controller failed its declared capability screen; all attempts are retained. Final calibration passed on eight new healthy validation clusters, 8/8 per law/bearing. A two-cluster fault screen was developmental. Core testing uses 24 new clusters (288 episodes); eight separate $\pm 45°$ clusters test angle transfer. Parameters, thresholds and stopping rules were frozen before both tests.

Warnings use only data through 400 ms: low mean command rate, an upper activity limit, and $|\log(S/\tilde{S}_H)|$ against law/side-specific healthy medians. A held-command forecast uses current geometry, neural turn command and pending burst; absent a trigger it assumes the earliest next-frame decision. A same-state actuator baseline sets future turn to zero. Both have exact simulated geometry, not implemented sensors or reachability guarantees. Lower fifth-percentile command/clearance and upper 95th-percentile log-score thresholds use all healthy validation records; the upper resource limit is $2\tilde{S}_H$. Healthy flag calibration does not guarantee task false-alarm rates.

All test records contribute sensitivity, specificity, false alarms and misses against contact through 1 s. Confidence intervals use 20,000 whole-seed percentile resamples, RNG 20261022. Controller sign-flip sensitivity assumes a symmetric cluster null, Holm-adjusted over two doses. Fixed 800/900/1000 ms task horizons are repeated observations, not new replicates. No test tuning, seed replacement or efficacy stopping is permitted. Inference remains conditional on one model, map and task.

IV. Independent Test Results

A. Fault Loss Depends on Feedback

All 96 healthy core episodes succeed: 24/24 for each law/bearing in Fig. 2. At 25% deletion, proportional control succeeds in 46/48 episodes and relay in 48/48; at 50%, these counts are 37/48 and 48/48. There are no healthy-to-fault gains. The 50% proportional losses are direction-dependent: 10/24 at $-30°$, 1/24 at $+30°$. Both healthy directions remain capable, avoiding the earlier directional floor.

The registered relay-minus-proportional net-loss contrast at 50% is -22.9 percentage points (pp), 95% cluster interval $[-33.3, -12.5]$, with 11/24 nonzero cluster contrasts.

TABLE I
FROZEN 400-MS WARNINGS ON ALL CORE RECORDS. TP/FN USE 13 COLLISIONS; FP USES 275 SUCCESSFUL TASKS. FULL CLUSTER INTERVALS, SPECIFICITY, FPR AND FNR ARE IN THE SUPPLEMENT. THESE ARE SCORES OF OUR RECORDS, NOT EXECUTIONS OF SPIKEFI OR SPIKINGJET.

Criterion	TP	FP	FN	Sens. (%)	FPR (%)
Command low	2	51	11	15.4	18.5
Activity upper	0	0	13	0.0	0.0
Activity two-sided	3	68	10	23.1	24.7
Actuator schedule	13	84	0	100.0	30.5
Held command	13	57	0	100.0	20.7

Symmetric-null sign-flip sensitivity gives raw $p = 0.000977$, Holm-adjusted $p = 0.00195$. At 25%, the contrast is -4.2 pp, interval $[-10.4, 0]$, adjusted $p = 0.50$; two nonzero clusters provide limited evidence. Relay's zero losses are observations, not zero-risk estimates. These new laws do not replicate the older six-cluster original/gated comparison, whose exploratory 25% $p = 0.25$ remains in the supplement.

A post-hoc trace diagnostic finds identical trigger times and pre-decision traces in all 144 matched law/scenario pairs. Steering is gated below the trigger threshold, so the laws first diverge after that decision. Median trigger time rises from 160 to 220 ms under 50% deletion for both laws; median absolute heading at the precontact 400-ms window is 38.7° for proportional control and 96.0° for relay. This describes a timing/turning pathway, not an identified mediation effect. Maximum turn rates match; executed turning effort does not. The result concerns feedback context rather than an intrinsically more fault-tolerant network.

B. Neural Criteria Miss Task Failures

All 400-ms assessments precede the earliest contact, 607.6 ms. Table I scores the same 288 records against 13 later collisions and 275 successful tasks. The command-rate criterion detects only 2/13 collisions; its sensitivity interval is $[0, 40.0]\%$. Two-sided activity detects 3/13 with more false alarms, and the upper activity limit detects none. Thus, simply adding a lower activity cutoff does not resolve the endpoint mismatch.

Both geometry-informed forecasts detect 13/13 collisions. Holding the current neural turn command reduces false alarms from 84 to 57 relative to the same-state zero-future-turn baseline. A secondary descriptive paired analysis gives a specificity gain of 9.8 pp, interval $[6.6, 13.4]$. Its remaining 20.7% false-alarm rate and exact-state information limit practical interpretation. Zero observed misses yield a degenerate bootstrap interval; the one-sided 95% bound on any miss in a new, identically defined seed cluster is 11.7%, not a conditional FNR guarantee. This candidate is not a validated protection algorithm.

C. Horizon and Exposure Have Distinct Effects

Only two collisions occur by 800 or 900 ms, versus 13 by 1 s. The 50% controller contrast is therefore -4.2 pp at the earlier horizons, versus -22.9 pp at 1 s. These are repeated observations, not extra samples; the declared horizon materially changes task loss.

In contrast, duration matching leaves every group median spike ratio unchanged. At 50% deletion, stopped/common medians are 1.505 for proportional control and 1.257 for relay; the maximum individual change over all pairs is 0.00325. The large historical stopped/common difference does not recur here. Proportional 50% activity has median ratio 0.981 at 400 ms and 1.505 over 1 s, illustrating distinct resource windows. Full-horizon post-contact continuation is not prospective warning evidence.

V. IMPLICATIONS AND LIMITS

The same functional deletion mask can receive different task-level fault labels under two healthy-capable feedback laws. A neural threshold and a task requirement should therefore be reported as separate specifications, with the actuator schedule and task horizon attached. The two-sided activity control shows that the discrepancy is not explained solely by choosing an upper envelope for a deletion fault.

Evidence remains conditional on one v783 graph, one assumed visual map and a simple finite-horizon pursuit task. The laws share actuator limits but use different turning histories; relay performance does not establish lower control cost or a hardware protection policy. Geometry-informed forecasts have more information than neural counts, and false alarms remain substantial. Sparse failure clusters and fixed calibration limit uncertainty estimates. Functional row deletion is uncalibrated to physical bit errors, and no energy, area, real-time execution or biological fault tolerance is established.

VI. CONCLUSION

An auditable paired assessment separates feedback-dependent task loss from neural-rate and activity criteria in a large connectome SNN. Independent evidence supports a 50% deletion contrast at the specified one-second endpoint; the 25% contrast remains uncertain and stopped-duration matching has little effect in the new cohort. The reusable result is a source-bound assessment and replay workflow with explicit task, timing and resource endpoints.

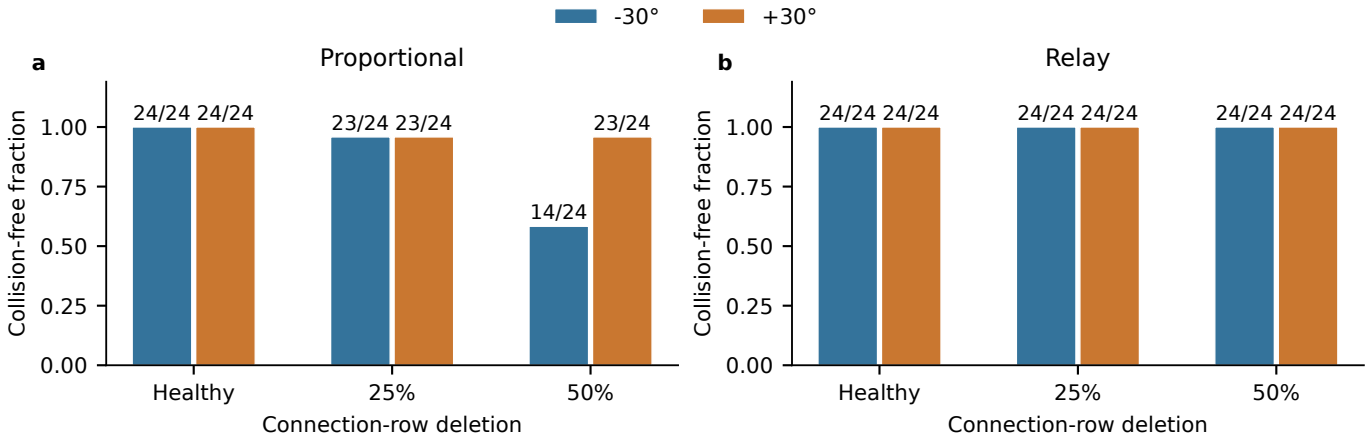


Fig. 2. All independent core outcomes at the fixed 1-s task horizon. Each bar contains 24 seed clusters; the two directions, laws and doses within a seed are paired, not independent replicates. Counts are raw successes/episodes. Both healthy directions succeed under both laws; 50% deletion produces different task loss despite the common mask and actuator limits.

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